

Neurohistology

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Neurohistology topics for IBB

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Light microscopy

- Neurons and neuroglia - in nissl and golgi stains
- Histology of major areas of the nervous system - spinal cord, cerebral cortex, cerebellar cortex, retina, internal ear

Ultrastructure

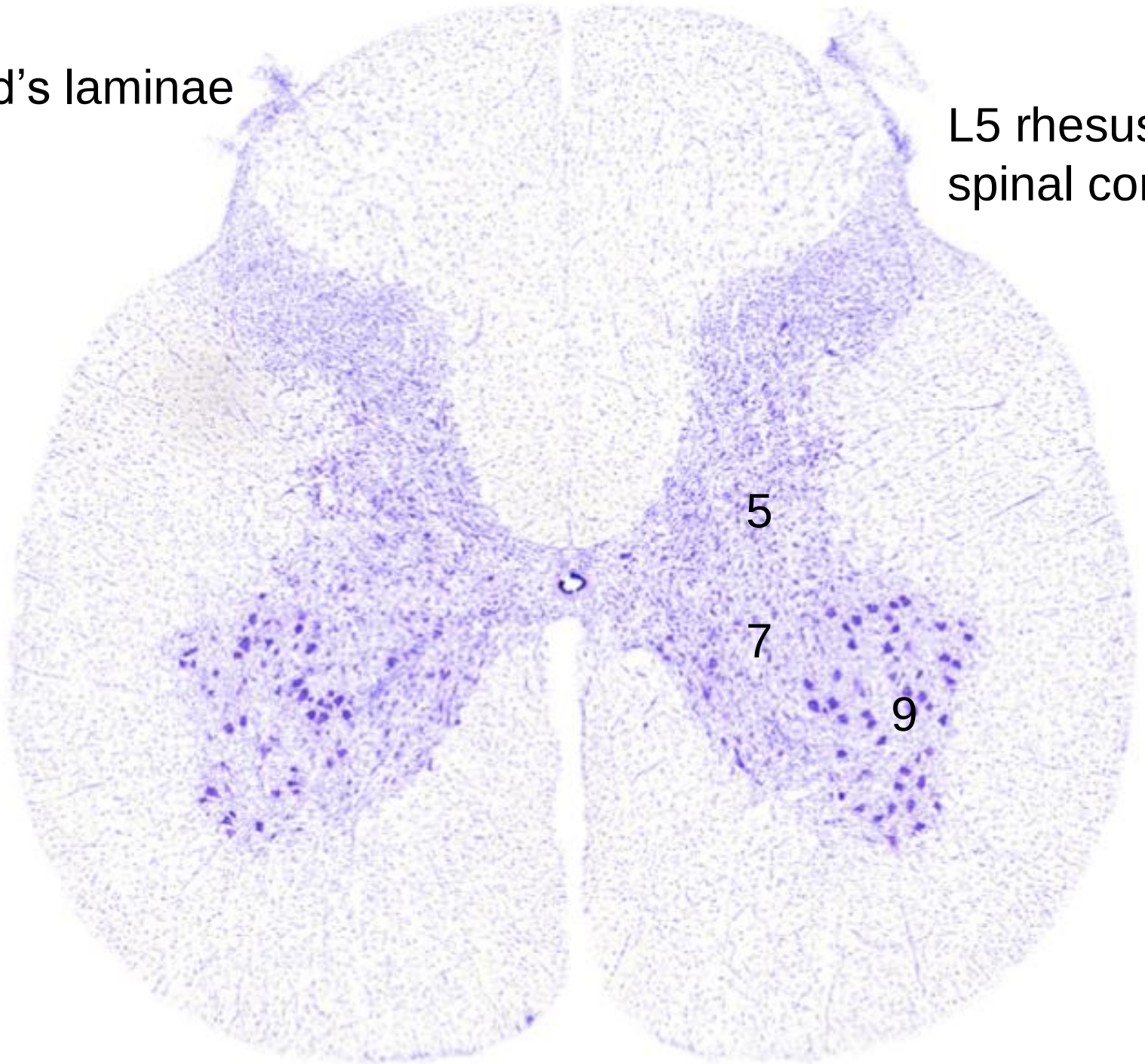
- Neurons (dendrites, dendritic spines, axons), glia, myelin sheaths. Schwann cells, oligodendrocytes
- Synapses - synaptic vesicles in excitatory and inhibitory synapses, dendritic spines, synaptic membrane thickenings in excitatory and inhibitory synapses

Spinal cord

- Six spinal cord regions
 - Two regions supplying the limbs – C5-T1 and L2-S1 – large ventral horn with prominent motor neurons
 - Two autonomic regions – T2-L1 (sympathetic) and S2-S4 (parasympathetic) featuring ChAT positive intermediolateral horn preganglionic neurons
 - Neck and tail regions – C1-C4 and S4-Co2 – neither a big ventral horn or an intermediolateral horn

Rexed's laminae

L5 rhesus
spinal cord

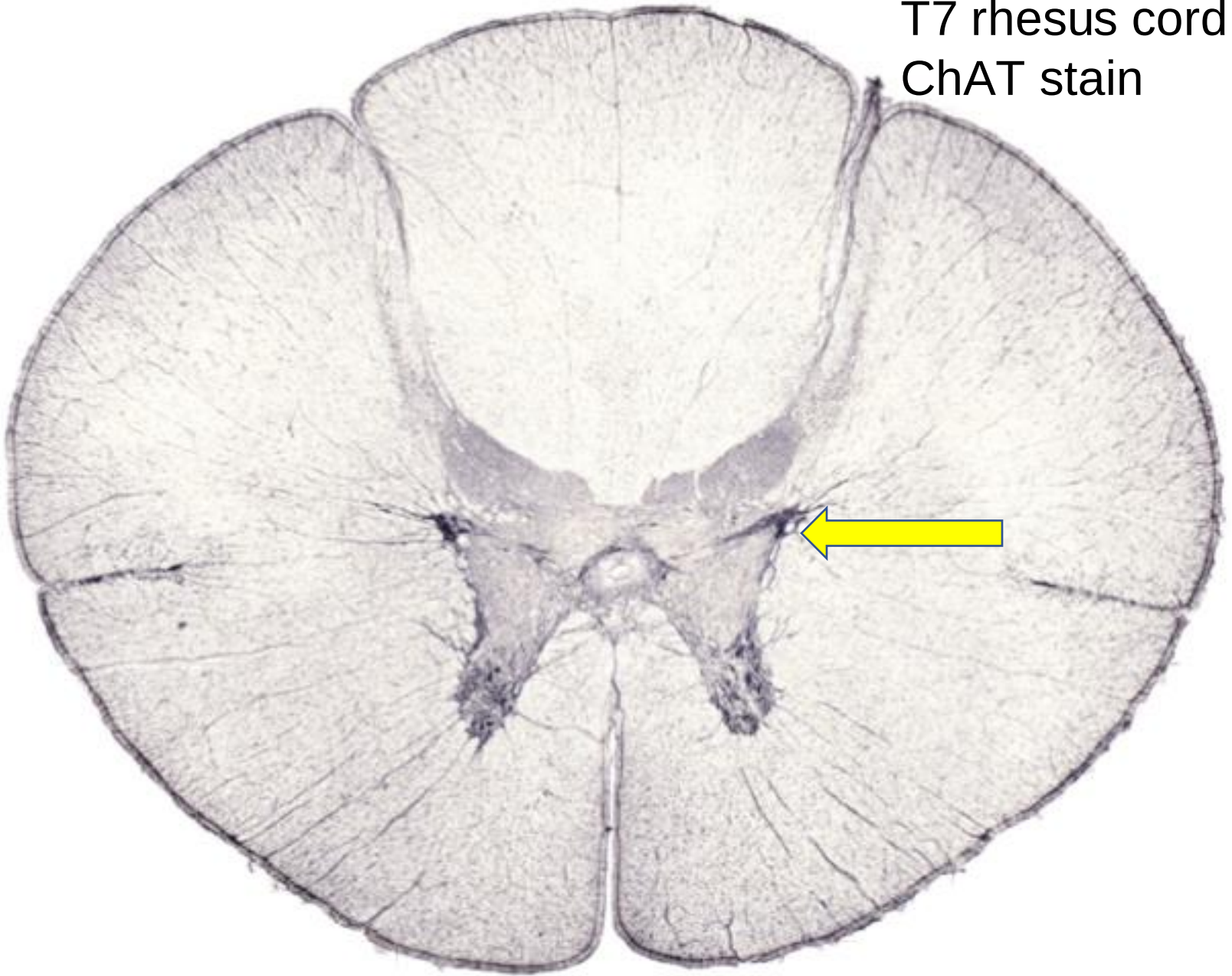


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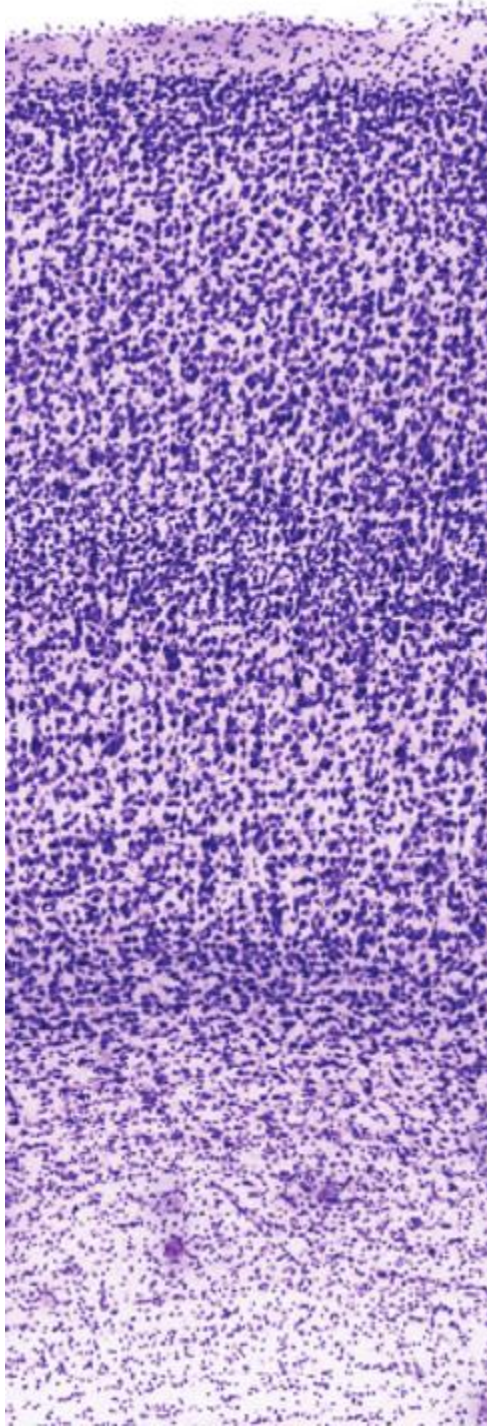
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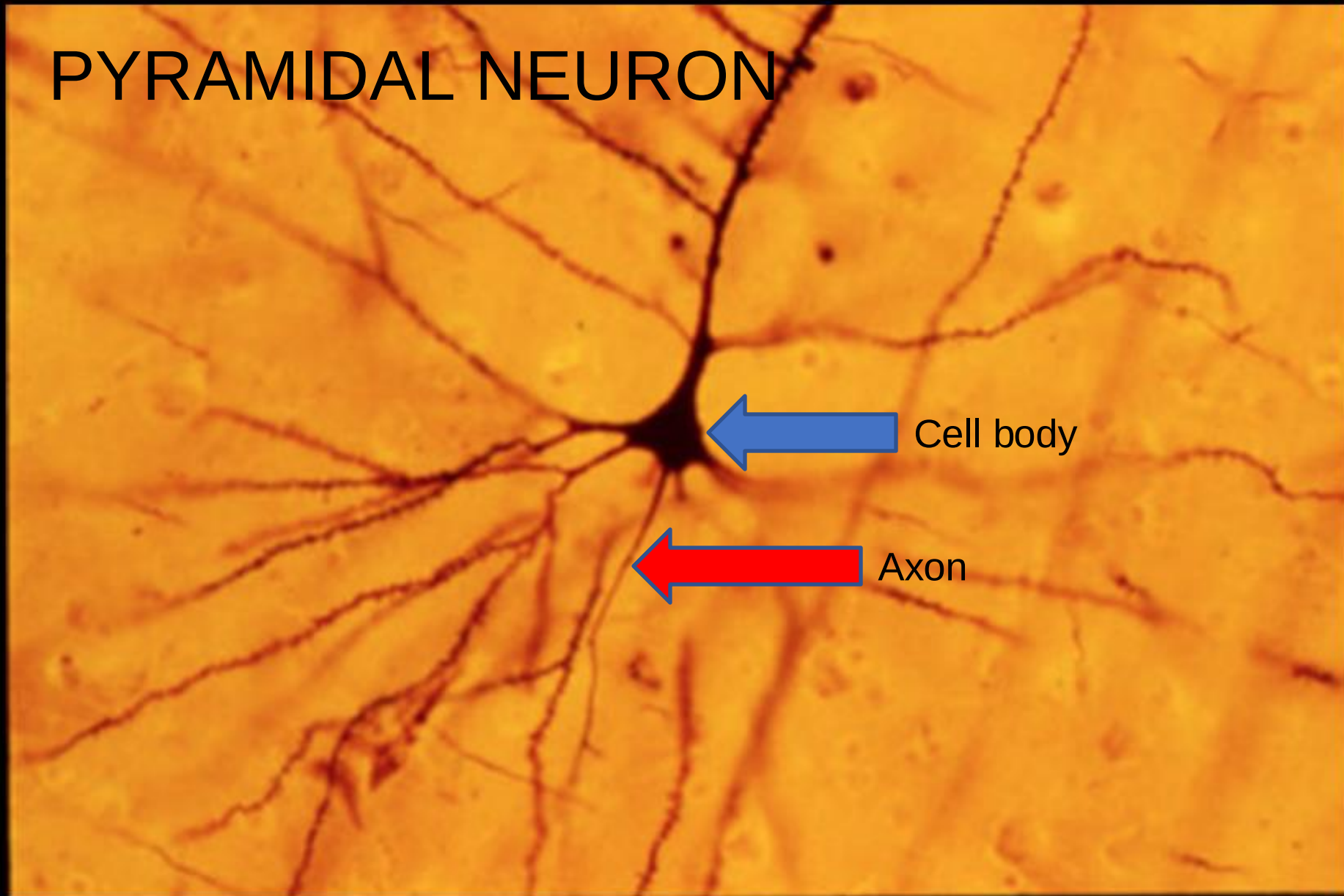
T7 rhesus cord
ChAT stain



Cerebral cortex and cerebellar cortex



PYRAMIDAL NEURON



HIPPOCAMPUS

Rhesus
hippocampus

Entorhinal
cortex

CA3

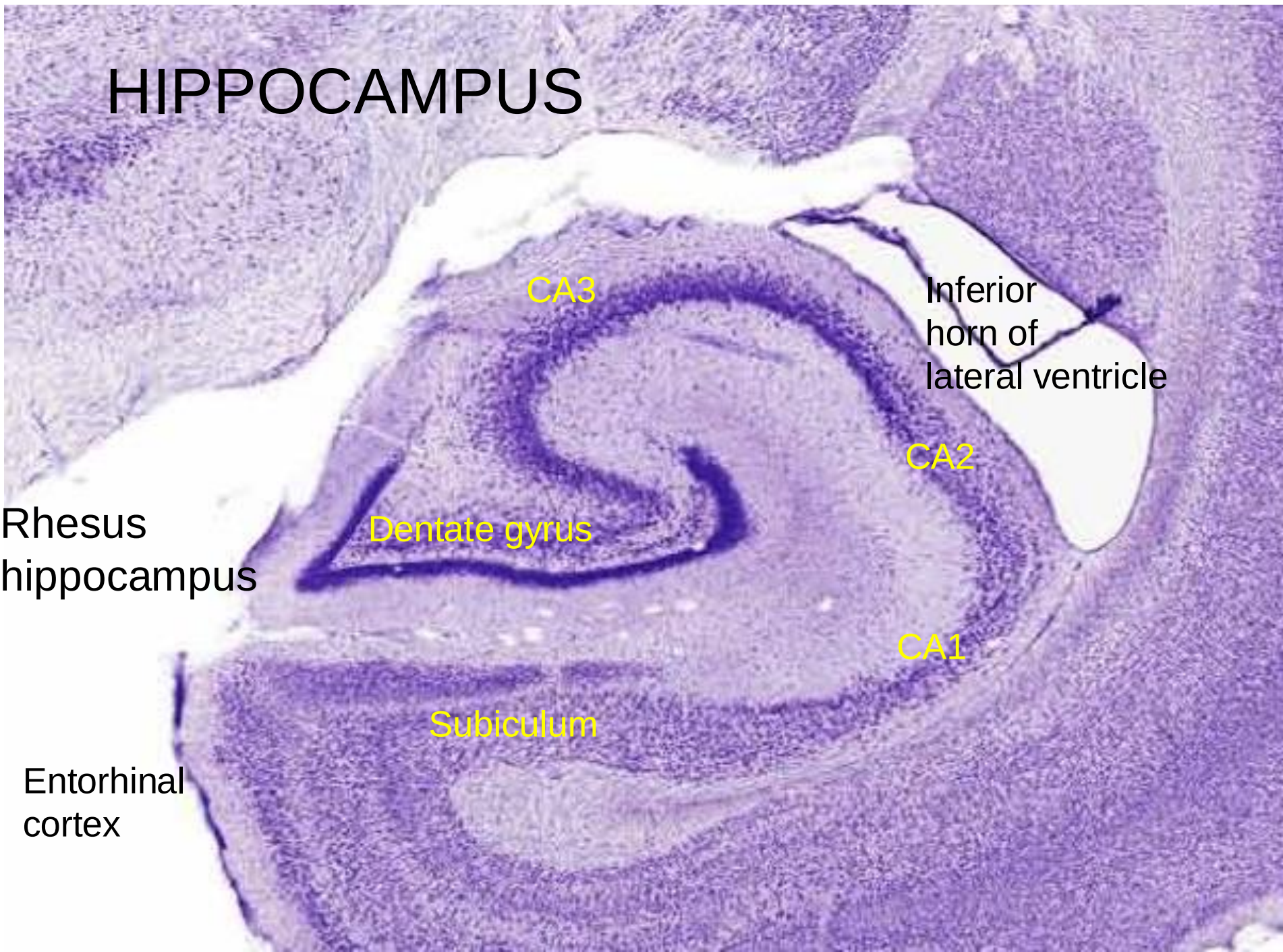
Inferior
horn of
lateral ventricle

CA2

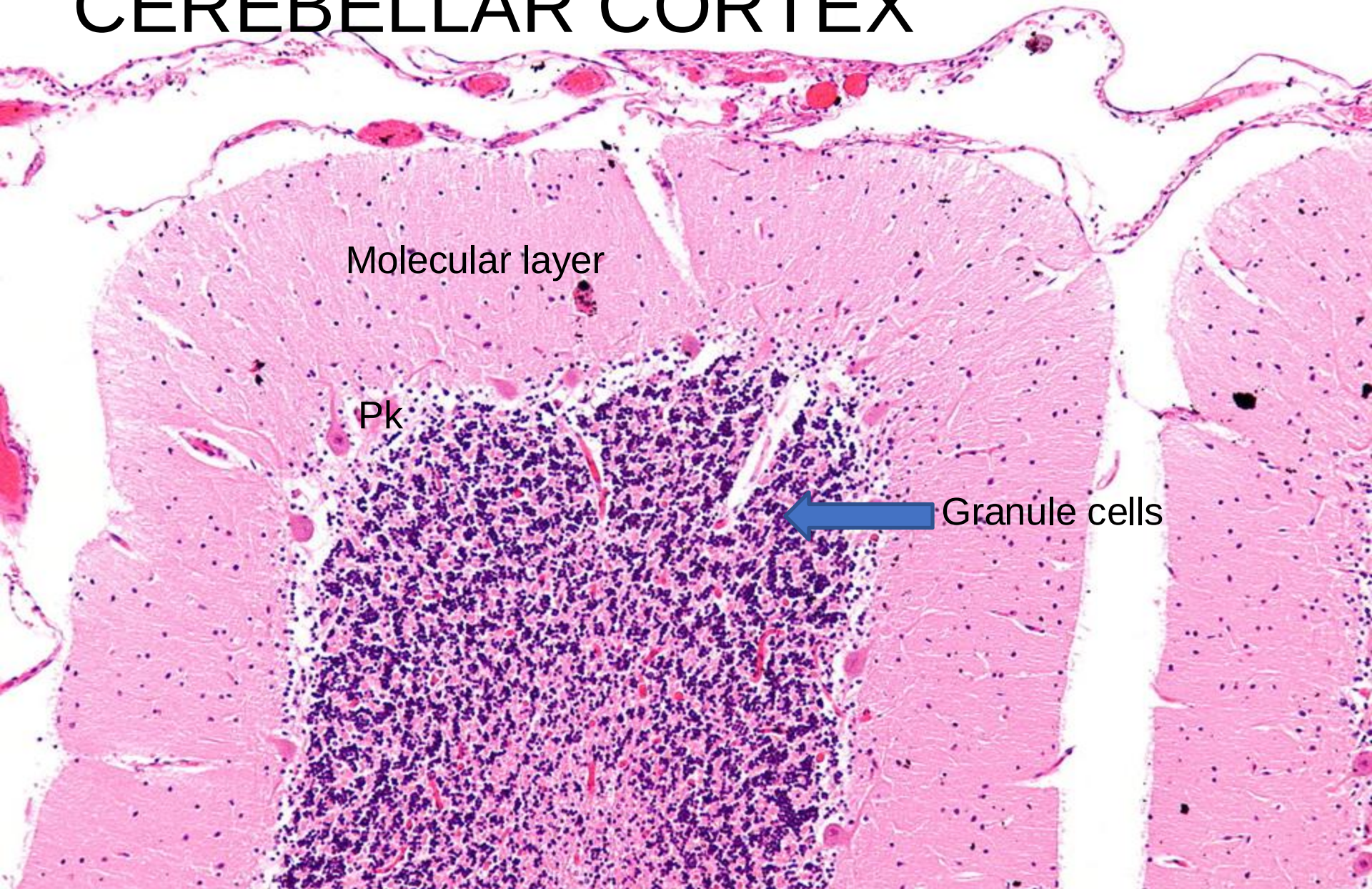
Dentate gyrus

CA1

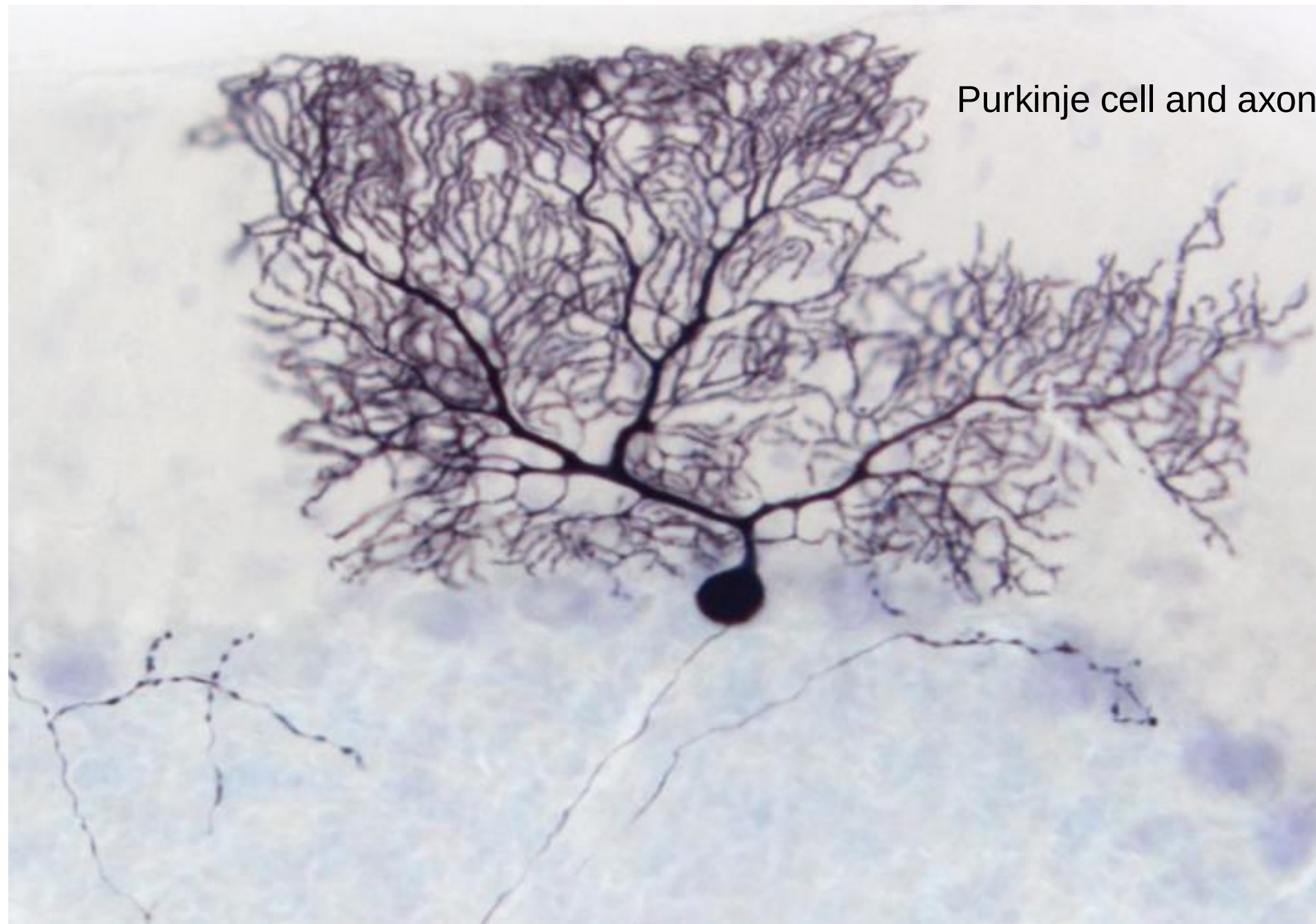
Subiculum



CEREBELLAR CORTEX



Purkinje cell and axon



Special sense organs

Retina

Vitreous layer

Ganglion cells

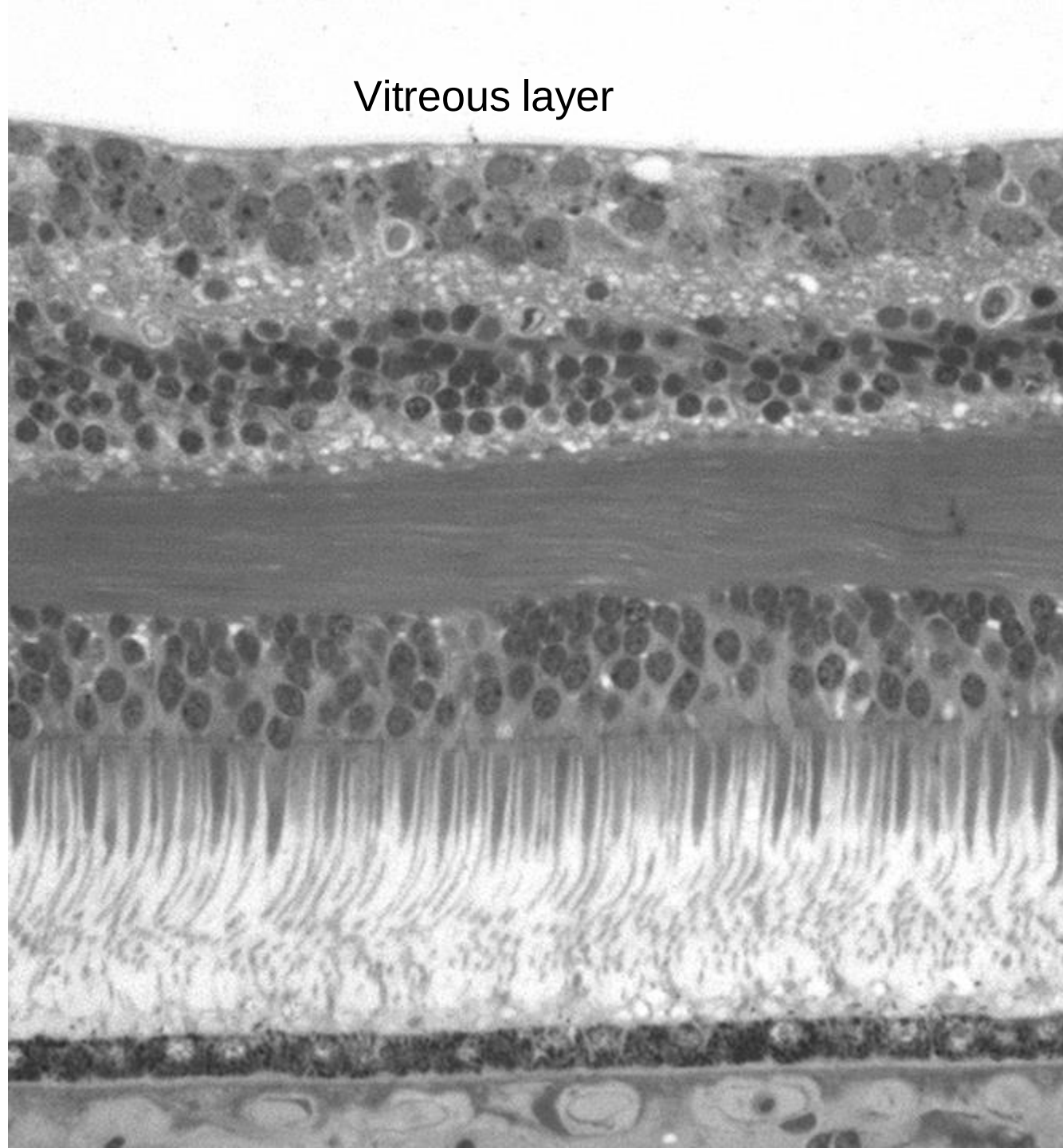
Bipolar cell nuclei

Rod and cone cell
nuclei

Rods and cones

Pigment cell layer

Choroid



Cochlea

Scala tympani

SPIRAL
GANGLION

Scala media

Organ of Corti

Scala vestibuli



Ultrastructure

Schwann cell



Axon

500 nm

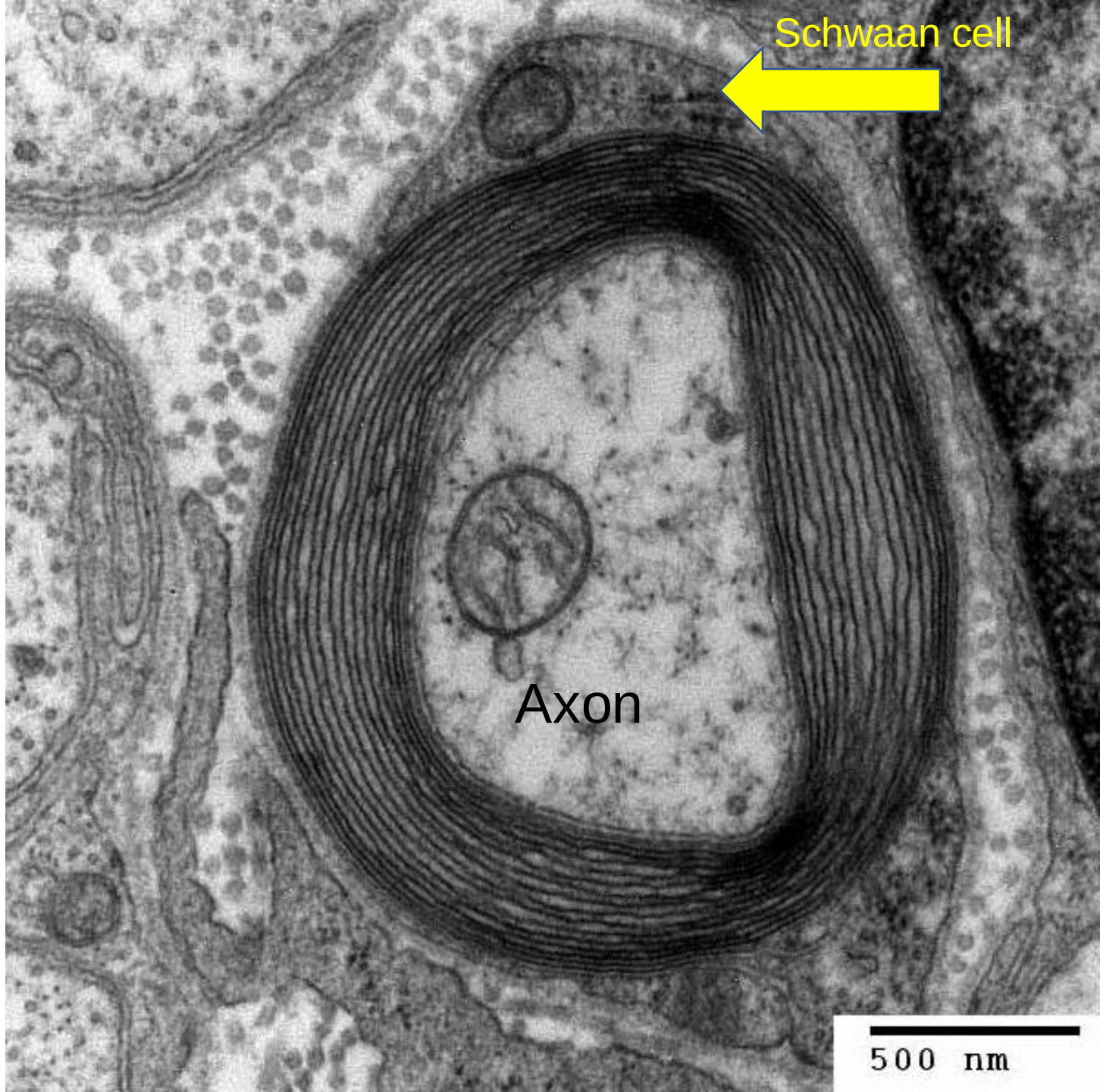


Image courtesy of Okinawa Institute
of Science and Technology

Bouton

Synaptic
vesicles

Post-synaptic thickening

Dendritic
spine

100 nm

